

Low-Cost Mechanical Ventilator Joaquín 1.0

Electronic Hardware and Embedded Software Support for Emergency Medical Equipment

Project Overview

This project focused on the development of electronic hardware and embedded software support for Joaquín 1.0, a low-cost mechanical ventilator developed in Guatemala during the COVID-19 emergency. The system was created as a response to the urgent need for respiratory support equipment during a period when access to commercial ventilators was limited.



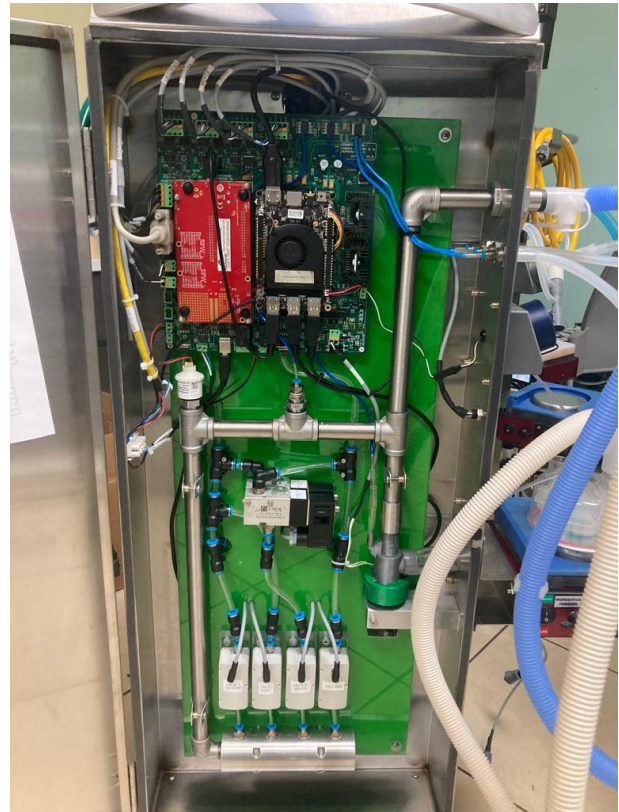
Figure 1. Joaquín 1.0 ventilator system overview.

Joaquín 1.0 was designed as a computerized mechanical ventilator for critical care environments with access to medical-grade compressed air and oxygen. The system included a graphical user interface, configurable ventilation parameters, alarm monitoring, a humidifier module, temperature sensing, battery backup, and patient respiratory circuits.

My work focused on the engineering side of the project, especially the electronic circuits used in the system, PCB verification, software support, and technical assistance during deployment.



(a)



(b)

Figure 2. (a) Main PCBs assembled and (b) Internal system of the ventilator.

Motivation

During the COVID-19 pandemic, many countries faced shortages of medical ventilators and respiratory support equipment. In Guatemala, this created a strong need for locally developed, lower-cost alternatives that could be manufactured, tested, and deployed using available technical resources.

The goal of the Joaquín 1.0 project was to contribute to an emergency medical device capable of supporting healthcare professionals during a critical period. From an engineering perspective, the project required reliable electronics, robust system integration, safety-oriented alarm behavior, and a user interface that doctors could understand and operate after training.

System Concept

Joaquín 1.0 integrated mechanical, electronic, and software subsystems into a single ventilator platform. The system included electronic control circuits, sensors, valves, alarms, a graphical display, communication between modules, and support for auxiliary components such as the humidifier and temperature sensing system.

The electronic hardware served as a bridge between the physical components of the ventilator and the control software. This required PCB-based control and interface circuits capable of reading sensor information, driving system outputs, supporting alarms, and communicating with the main control interface.

My Role

My contributions to this project included:

- Designed electronic circuits used in the ventilator system.
- Supported PCB design, component selection, and hardware implementation.
- Verified manufactured PCBs and checked their functionality before system integration.
- Helped develop and test software used to support the operation of the ventilator system.
- Assisted doctors and medical personnel during training sessions so they could understand the system interface and operation.
- Collaborated with a multidisciplinary team working under emergency-response conditions.

Methodology

The development process began by identifying the electronic requirements of the ventilator system, including sensor interfaces, control outputs, alarm behavior, power-related signals, and communication between hardware modules. Based on these requirements, electronic circuits were designed and implemented into custom PCBs. The boards were then reviewed, manufactured, assembled, and electrically verified before being integrated into the complete ventilator system.

After hardware verification, the system was tested together with the software and mechanical components. The development process involved debugging electronic circuits, checking PCB behavior, supporting software implementation, and validating system operation. Once units were

prepared, technical assistance was provided during delivery and training so that doctors could become familiar with the interface and general system workflow.

Key Technical Areas

This project demonstrates experience in:

- PCB design and validation
- Embedded systems
- Device prototyping
- Multidisciplinary engineering collaboration
- Technical training and deployment support

Results and Demonstrations

The project contributed to the development of a functional low-cost mechanical ventilator platform during the COVID-19 emergency. 30 ventilators were manufactured and donated to public hospitals. The ventilators were used and some of them are still being used to this day.



Figure 3. Ventilator being used by doctors.



Figure 4. Ventilator being delivered to a public hospital.

Engineering Challenges

This project presented several important engineering challenges. The system had to be developed during an emergency period, which required fast design iterations, practical component selection, and close collaboration between engineering and medical personnel. The electronics needed to be reliable enough for system-level testing and easy to integrate with the rest of the ventilator platform.

Confidentiality and Safety Note

This portfolio summary describes the project from an engineering and system-development perspective. It does not provide medical instructions, clinical recommendations, or operating procedures. Some technical details, design files, firmware, and implementation information are omitted or generalized due to confidentiality and safety considerations.

Project Status

This project was developed during my work at Ingenio La Unión as part of the emergency response to the COVID-19 pandemic in Guatemala. My contribution focused on electronic circuit design, PCB verification, embedded software support, system testing, ventilator delivery, and technical assistance during doctor training. A total of 30 ventilators were manufactured and donated to public hospitals, and some units continued to be used after deployment.